

KDD (Knowledge Disc in Databases)

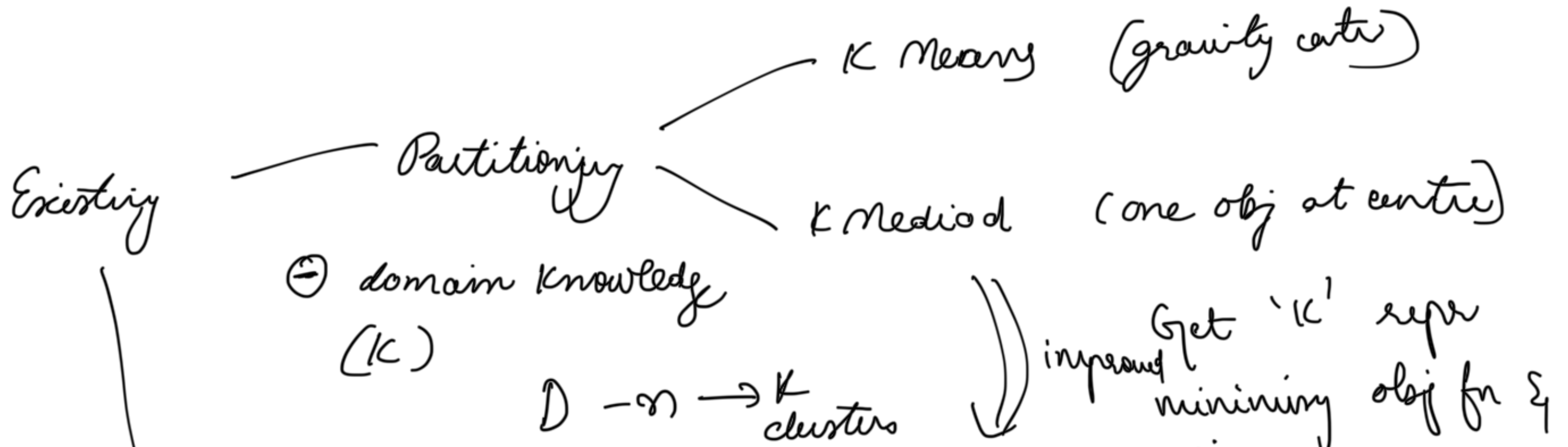
domain
knowledge

arbitrary
shape

large database

DBScan - Density Based Spatial Clustering in applications
w/ noise

- Class identification



⊖ convex shapes

Hierarchical

↙
Dendrogram

$D \rightarrow$ subsets iteratively

⊖ pick $D_{\min}$

termination
condition

Ejcluster

- walk for points a to b
in small steps

⊕ non convex

⊖ $O(n^2)$

Clara

$K: 2 \rightarrow n$

Max silhouette
coefficient

$O(n)$ calls

⊖ memory
large db

⌋ improve

On relevant
part of DB

assign

↘
Voronoi diagram

Convex shapes
only

$$\{x \mid \|x - \mu_i\| \leq \|x - \mu_j\| \mid i \neq j\}$$

↙
half space

→ Density based notion of Clusters

def

$$1. N_{\epsilon}(p) = \{q \in D \mid \text{dist}(p, q) \leq \epsilon\}$$

naive: for each point $\in C$, need minPts but border pts usually have lesser neighbours, will have to pick $\vee$ small minPts

def 2 Directly density reachable

p is ddr for q if

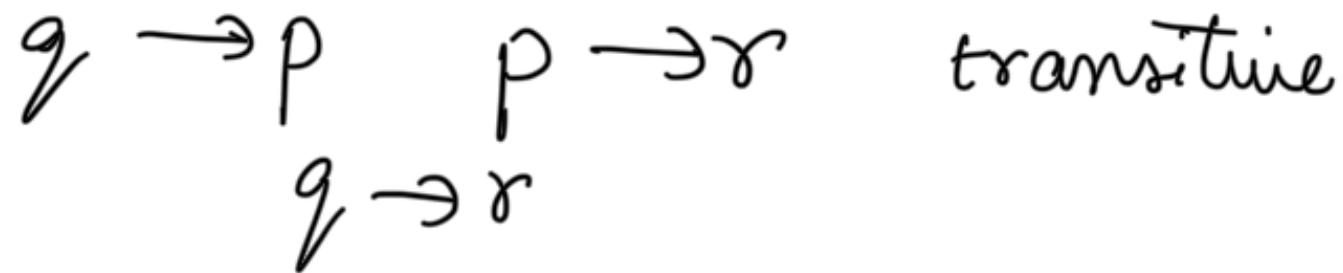
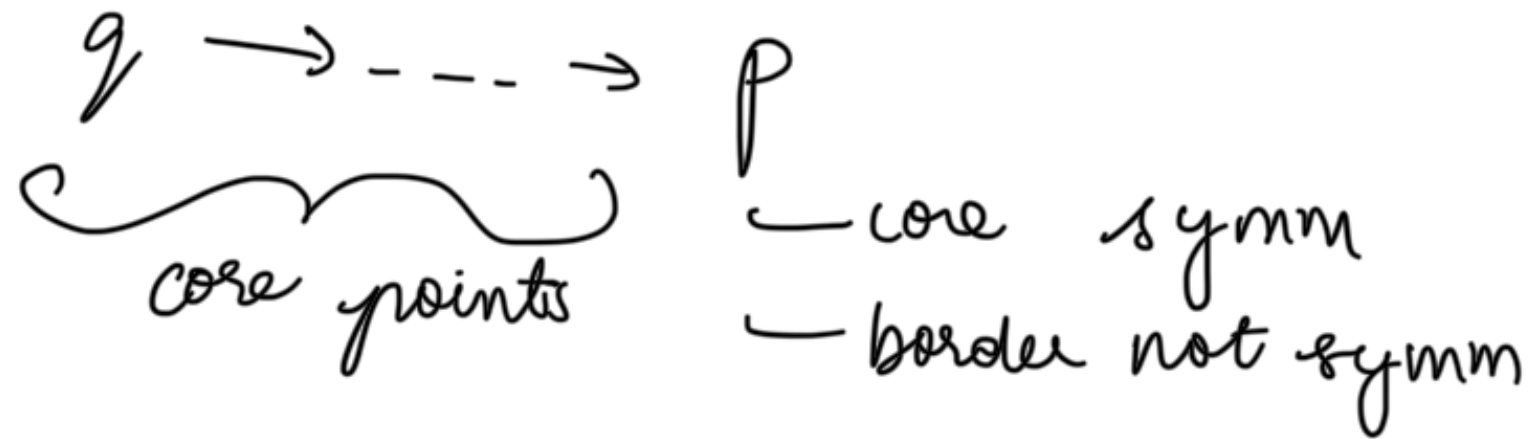
$q \rightarrow p$
 $\underbrace{q}_{\text{core}}$

$|N_{\epsilon}(q)| \geq \text{minPts}$

not symmetry for
 $q \rightarrow p$
 $\underbrace{q}_{\text{core}} \quad \underbrace{p}_{\text{border}}$

def 3

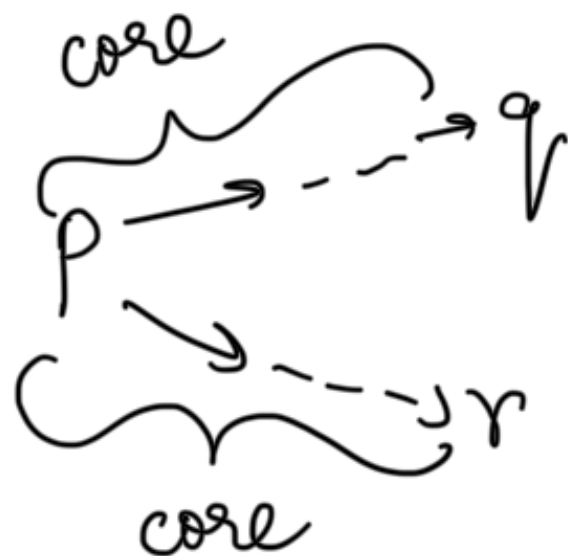
density reachable



$\forall$ border pt q , $\exists$ core point p s.t. q is dr from p

def 4

Density Connected



q, r : density connected
Symmetric

def 5 Cluster

C : maximal set of density connected points
 at least 1 cluster
 $|C| \geq 1$ min pts
 non empty

1) $\forall p, q, p \in C, q \in C$
 $p \rightarrow \dots \rightarrow q$
 core
 dr

(maximality)

2) $\forall p, q \in C, p, q$ density connected (connectivity)

def 6 (noise)

$$= \{p \in D \mid \forall i : p \notin C_i\}$$

Lemma 1

p -core
 $O = \{o \mid o \in D, p \rightarrow \dots \rightarrow o\}$ is a cluster w.r.t ϵ_{ps} & MinPts

Lemma 2

$$C = \bigcup$$

so take any core point p , C is set of all points density reachable from p

1) $O \subseteq C$

$O = \{ q \mid p \rightarrow \dots \rightarrow q \} = \text{density reachable from } p$
 By definition of Cluster
 (maximality)

2) $C \subseteq O$

let q be a point in the cluster

p, q are density connected so



960

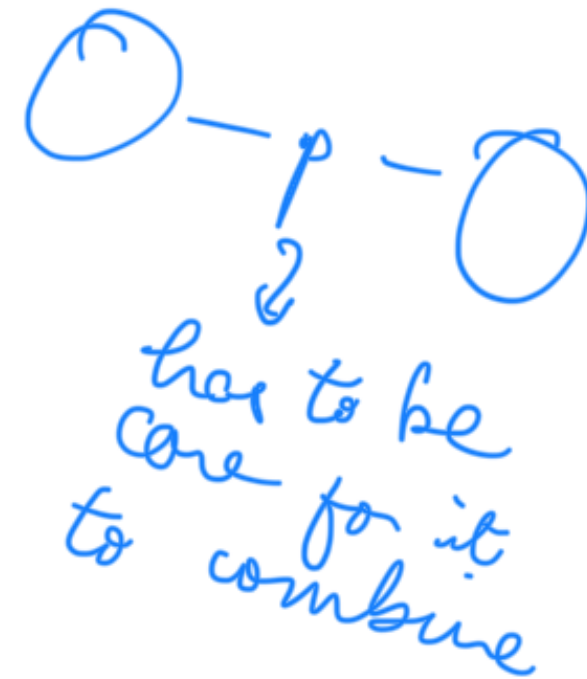
reachable
from p

$\Rightarrow$ So "critical" edges taken care of? Need

$\Rightarrow$ Unique answer for minPts & ϵ s

K-Means depends on initial
cluster centres

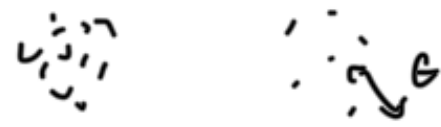
inc $\epsilon \uparrow$ merge clusters, absort noise
minPts $\downarrow$ more "core" merge



$\Rightarrow$ Cluster

Pick one global ϵ and MinPts to catch thinnest cluster.

w/o merging / swallowing
noise?



Threshold
corresponding
to the small
city you still want
to detect

$$C_i \quad f_i \approx \frac{\min P_{te}}{V(\epsilon p_s)}$$

$$f_i \propto \frac{1}{\epsilon p_s^d}$$

DBScan (Set of Pts, ϵ , minPts)

ClusterId := nextId(Noise) (initialize)

↙ for point in pts

$O(n)$

if point unclassified

if ExpandCluster ()

ClusterId := nextId(ClusterId)

if core returns true

ExpandCluster () :

seeds = pts.regionQuery(pt, E)

if |seeds| < minPts

pts.changeCId (Point, Noise)
ret F

else

Seeds $\rightarrow$ Cluster ID

seeds.delete(Point)

while seeds

curr = seeds[0]

next = ... (noise)

range query
 $\log n$

$O(n \log n)$

res = VUGPs (...)

if $|res| \geq \text{minpts}$

for $resp$ in res

if $\in \{\text{Unclassified Noise}\}$

if unclassified

seeds.append(resp)

delete Curve)

→ K-dist CD)

pt → distance from k^{th} nearest neighbour

4-dist (sorted K-dist graph)

core points

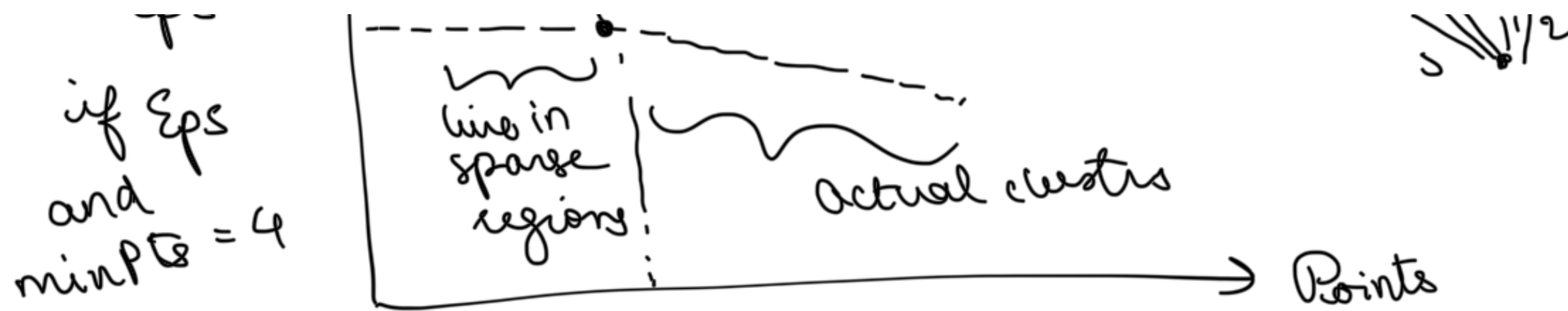
neighbour

unless at line

$5\text{-dist} \approx 4\text{-dist}$

↓
best number

$s_i / 4 \cdot i$



$\Rightarrow$ $\ominus$ variable densities

$\ominus$ can't automate

$\ominus$ high dimensional data

param selection? multiple elbows?

$\ominus$ border points

ambiguous?

all pts

$\approx$ ^vequally far apart